VGIC
VEHICLE-GRID INTEGRATION COUNCIL

March 3, 2026

CPUC Energy Division Tariff Unit
505 Van Ness Avenue
San Francisco, California 94102
EDTariffUnit@cpuc.ca.gov

Re: Comments of the Vehicle-Grid Integration Council on Draft Resolution E-5452. Southern California Edison Orchestrated Charging and Advanced Resiliency for Distribution Vehicle Grid Integration Proposal Using Low Carbon Fuel Standard Holdback Revenue

Dear Sir or Madam:

Pursuant to the provisions of General Order 96-B the Comment Letter accompanying Draft Resolution E-5452, the Vehicle-Grid Integration Council ("VGIC") respectfully comments on the proposed approval with modifications of Southern California Edison's ("SCE") Advice Letter 5536-E, *Southern California Edison's Request for Approval to Update Its Low Carbon Fuel Standard Holdback Revenue Return Implementation Plan to Include a New Vehicle Grid Integration Program and to Qualify for Exemption from Public Utilities Code Section 851*("Advice Letter").

I. INTRODUCTION.

VGIC is a 501(c)6 membership-based advocacy group committed to advancing the role of vehicle-grid integration ("VGI") through policy development, education, outreach, and research. VGIC supports the transition to a decarbonized transportation and electric sector by ensuring that the value of flexible EV charging and discharging is recognized and compensated, thereby achieving a more reliable, affordable, and efficient electric grid.

VGIC strongly supports the Draft Resolution's approval of the Orchestrated Charging and Advanced Resiliency for Distribution ("ORCHARD") program's managed charging component. The Commission correctly finds that orchestrated load management aligns with Decision 20-12-027 and with the California Air Resources Board's updated Low Carbon Fuel Standard ("LCFS") regulations, which identify managed charging and other VGI initiatives as appropriate uses of holdback funds. Approving the managed charging portion of ORCHARD represents an important step toward achieving the goals articulated in SB 676 to maximize feasible and cost-effective VGI by 2030.

The managed charging component of ORCHARD addresses a real and growing grid need. As EV adoption accelerates, secondary distribution peaks driven by synchronized time-of-use ("TOU") charging behavior are becoming more pronounced. Static price signals alone are insufficient to mitigate localized transformer and circuit constraints. By layering dynamic, circuit-level coordination, ORCHARD can more precisely shape load, defer distribution upgrades, and improve overall system efficiency. This type of localized, utility-orchestrated load management is exactly the kind of scalable demand flexibility resource contemplated in the Commission's transportation electrification and integrated resource planning proceedings.

While VGIC wholeheartedly supports approval of the managed charging portion of ORCHARD, **the Draft Resolution errs in rejecting SCE's proposal to provide targeted incentives for grid-parallel bidirectional charging systems.** As discussed below, bidirectional functionality unlocks additional use cases beyond backup power and is fully consistent with the Commission's VGI objectives and LCFS policy framework.

II. THE DRAFT RESOLUTION ERRS BY REJECTING THE BIDIRECTIONAL CHARGING SYSTEM INCENTIVE PROPOSAL.

As discussed above, VGIC strongly supports approval of ORCHARD′s managed charging component because it enables more precise, circuit-level load coordination and advances the Commission′s transportation electrification and reliability objectives. However, the Draft Resolution treats managed charging and bidirectional functionality as distinct, unrelated program elements rather than complementary components of a comprehensive VGI strategy. If managed charging allows the utility to shape when EVs consume electricity, bidirectional charging allows EVs to become active grid resources capable of both reducing load and supplying energy during constrained periods. In this way, promoting bidirectional charging system adoption unlocks deeper response per EV as, for example, a customer can move from simply avoiding 10 kW of load to avoiding 10 kW of load *and* adding 10 kW to the grid, effectively doubling their contribution.

The Draft Resolution rejects SCE′s proposal to provide incentives for the deployment of grid-parallel, or Rule 21-interconnected bidirectional charging systems. SCE′s proposal included an $800 rebate for bidirectional charging equipment, a number based on the cost of the Rule 21 interconnection application fee. Additional incentives were also proposed for income-qualified customers, up to $7,800. The Draft Resolution states that SCE′s proposal for bidirectional charging equipment incentives is "not sufficiently justified," largely because a Rule 21 interconnection is not required to use bidirectional charging systems as solely backup power resources.[1]

This reasoning overlooks two critical points. First, ORCHARD is explicitly designed to deliver grid and ratepayer value, not merely resiliency benefits for individual customers. There are several existing and pending export programs and rates that customers with this equipment could participate in to support the grid. Second, grid-parallel bidirectional functionality unlocks incremental reliability and system benefits that go far beyond what managed charging alone can achieve. SCE's proposal is appropriately structured to overcome one of the largest barriers to unlocking those broader grid services: the upfront costs of purchasing, installing, and interconnecting these systems.

A. Bidirectional charging can provide significant value for California′s reliability and ratepayers.

[1] Draft Resolution at 15.

There is significant value to unlocking grid support value from EVs beyond managed charging and load reduction. While managed charging can align EV charging load with real-time grid conditions, the maximum EV charging load reduction is fundamentally limited by a customer's initial plan to charge during a given period. However, export opportunities can elicit exports from EVs that may not have otherwise planned to charge at all during that period. In other words, the maximum EV charging export capability is not limited by a customer's initial plan to charge during a given period, enabling deeper peak system load reductions – in concept, *greater* than twice as much net peak load reduction.

To provide a more specific example, consider an EV charging at 4 kW during a local or system peak period. Managing charging during that window could induce the EV to reduce charging to 0 kW. This achieves a corresponding reduction of 4 kW at the local and system level. If exports were encouraged during that same period, the EV could stop charging and export power back to the grid at, for example 10 kW. Through the addition of exports, this EV provides a total peak load reduction of 14 kW (i.e., 4 kW load reduction + 10 kW grid export).

California has already built an immense catalogue of studies and modeling results showing the value of these incremental megawatts:

- **CPUC, Integrated Resource Planning Assumptions:** The Commission's Integrated Resource Planning (IRP) modeling estimates that over 13,500 MW of additional capacity could be provided by bidirectional charging systems in 2045.[2] This is incremental capacity beyond what would be offered by managed charging.

- **PG&E, Electrification Impact Study Part 2:** PG&E's Electrification Impact Study Part 2 found that 859 MW of additional demand flexibility could be unlocked through bidirectional EV exports.[3] Importantly, this is also fully incremental to the flexibility associated with active managed charging.

- **Union of Concerned Scientists, "Harnessing the Power of Electric Vehicles":** A June 2025 California-specific study by the Union of Concerned Scientists shows that, while unidirectional managed charging can save $4.2 billion of system costs by 2045, bidirectional charging unlocks an additional $3.5 billion even at a modest 12.5% participation rate in bidirectional charging activities (and a 50% participation rate in unidirectional charging activities). Increasing bidirectional charging activities to 25% enrollment yields a 60%

[2] California Public Utilities Commission ("CPUC"), Draft Inputs & Assumptions, 2024 – 2026 Integrated Resource Planning (IRP) at p.111. https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energy-division/documents/integrated-resource-plan-and-long-term-procurement-plan-irp-ltpp/2024-2026-irp-cycle-events-and-materials/2025_draft_inputs_and_assumptions_doc_20250220.pdf
[3] PG&E EIS Part 2 at 34.

increase in benefits compared to exclusively unidirectional charging future.[4]

- **Stanford University, "Residential vehicle-to-grid profits under dynamic pricing: The role of real-world charging behavior":** This study uses real-world EV charging behavior data from Volkswagen Group and PG&E's real-world dynamic export rate pilot price signals to assess the opportunity for bidirectional charging customers to support the grid. This study is rich with relevant findings, including those quoted below. Notably, the study finds that supporting at-home bidirectional charging is more beneficial than "V2G everywhere." Additionally, because the study is based on PG&E's hourly flex pricing, which closely reflects real-time grid conditions and is based only on marginal costs, **all modeled customer revenue results from customer response that, by definition, yields net positive benefits for the grid.** In other words, the study finds that EV owners can earn an annual *profit* of $1,181 per EV by exporting *only* when it is **most beneficial to the grid**.
    - "…if managed charging is allowed with dynamic pricing, shown in Fig. A.3, savings increase but are limited by lack of energy export."
    - "V2G provides savings compared to baseline charging for all circuits over the entire year."
    - "As this scenario also assumes V2G-capable chargers are installed at every public charging station, thus incurring additional installation costs, V2H Home is overall more economically viable."[5]

- **EV.Energy and Brattle, "The Utility Playbook: Turning EV Grid Risk into a $30 Billion Opportunity":** This study leverages real-world data from California, New York, Colorado, and PJM to inform a U.S.-wide estimate of the avoided costs associated with VGI by 2035. The study finds ~$5 billion of avoided costs from passive managed charging, ~$13 billion from active managed charging, and ~$12 billion from bidirectional charging. This equates to $330-$1,320 per bidirectional EV per year, with California likely seeing the upper end of that spectrum due to its higher costs relative to the national average.[6]

- **Electric Power Research Institute (EPRI), "Distribution System Constrained Vehicle-to-Grid Services for Improved Grid Stability and Reliability":** This California-specific value analysis conducted by the EPRI project team, supported by E3 and several automakers, shows a cumulative maximum benefit to the grid of bidirectional charging (net of cost

[4] Samantha Houston, David Reichmuth, and Mark Specht. *Harnessing the Power of Electric Vehicles: Vehicle-Grid Integration for a Cleaner, Cheaper, More Reliable California Electricity System.* June 2025. Union of Concerned Scientists. https://doi.org/10.47923/2025.15888. Accessed January 13, 2026.
[5] Sonia Martin, William A. Paxton, and Ram Rajagopal. *Residential vehicle-to-grid profits under dynamic pricing: The role of real-world charging behavior.* March 2026. Advances in Applied Energy. https://www.sciencedirect.com/science/article/abs/pii/S0360544225024156?via%3Dihub
[6] William Goldsmith et al., *The Utility Playbook: Turning EV Grid Risk into a $30 Billion Opportunity*. 2025. EV.Energy and Brattle. https://www.ev.energy/en-us/resources/value-of-managed-charging

increment) ranging between $450/year per vehicle to $1,850/year per vehicle. Since ratepayers are being required to provide some support for the public PEV charging infrastructure, some of these benefits can defray or defer some of the infrastructure upgrade costs, thereby limiting upward electricity rate pressure.[7]

- **Lawrence Livermore National Laboratory, "The Potential Electric Grid Benefits of Vehicle-to-Grid Technology in California":** This study demonstrates that incremental benefits for California's grid, meaning the utility's costs and benefits of serving and managing EV charging, equal $337-$1,472 annually per EV for "V2G vs. V1G."[8]

- **UC Irvine, "The value of consumer acceptance of controlled electric vehicle charging in a decarbonizing grid: The case of California":** This California-specific study finds, "that increased participation in smart charging and vehicle-to-grid increases zero-carbon generation uptake by up to 5.2% and 11.1%, respectively. The value of smart charging only reaches $87 per vehicle-year while that for vehicle-to-grid can reach $2,850 per vehicle-year."[9]

- **Pacific Northwest National Laboratory, "Assessing the Energy Equity Benefits of Mobile Energy Storage Solutions":** This study looks at the value of bidirectional charging systems, including personal light-duty vehicles, school buses, and freight trucks, finding that EVs can provide at least 3 hours of critical load service not only for onsite loads, but for loads on the entire feeder.[10]

B. There are a variety of non-backup power bidirectional charging use cases that SCE customers can participate in today.

The Draft Resolution rejects the bidirectional equipment incentive largely because it envisions the only use case for bidirectional charging systems being backup power. The Draft Resolution states that most bidirectional charging systems sold at the time of SCE's Advice Letter submittal used their bidirectional discharge solely for backup power.[11] The Draft Resolution then outlines that Rule 21 interconnection is not required for bidirectional charging systems being used solely for backup power in isolated operation mode, often referred to as a "break-before-make" transfer scheme. It is

[7] Sunil Chhaya, PhD, et al. *Distribution System Constrained Vehicle-to-Grid Services for Improved Grid Stability and Reliability.* CEC. March 2019. https://www.energy.ca.gov/sites/default/files/2021-06/CEC-500-2019-027.pdf
[8] Jonathan Donadee et al. *Potential Benefits of Vehicle-to-Grid Technology in California: High Value for Capabilities Beyond One-Way Managed Charging*. Lawrence Livermore National Laboratory. IEEE Electrification Magazine. June 2019. https://www.osti.gov/pages/biblio/1557041
[9] Brian Tarroja and Eric Hittinger. *The value of consumer acceptance of controlled electric vehicle charging in a decarbonizing grid: The case of California.* UC Irvine. Energy (Volume 229). https://www.sciencedirect.com/science/article/pii/S0360544221009397
[10] Jessica Kerby, et al. *Assessing the Energy Equity Benefits of Mobile Energy Storage Solutions* 6th E-Mobility Power System Integration Symposium October 2022. https://www.pnnl.gov/sites/default/files/media/file/Assessing_the_energy_equity_benefits_of_mobile_energy_storage_solutions.pdf
[11] Draft Resolution at 15.

true that these systems (i.e., those used *solely* for backup power in isolated operation mode) are not subject Rule 21 interconnection requirements.

However, backup power use cases provide limited grid benefits. There are other use cases for customers to use bidirectional charging equipment that provide the grid and ratepayer value described above, including:

- **Bill Savings and Peak Shaving**: As discussed by SCE, "Even without export tariffs, V2X technology can provide grid benefits and cost-savings to individual customers by performing whole home peak shaving during existing TOU peak periods."[12]SCE estimates that customers could save $500 a year from peak shaving alone.[13] For example:
    - In October 2025, Ford stated: "Now, with [Ford Home Power Management](https://www.fromtheroad.ford.com/us/en/articles/2025/introducing-ford-home-power-management), in select markets where electricity rates change throughout the day, customers that purchase the Ford Charge Station Pro, Home Integration System, Home Backup Power software, and receive the Home Power Management software enhancement can **charge their F-150 Lightning with lower cost electricity — often overnight during off-peak hours — and then use that stored energy to power their home at a later time when grid electricity rates are higher during peak hours.**"[14]
    - In November 2025, Volvo launched its bidirectional charging partnership with dcbel. The offering website states: "Powerful. Even when parked. Unlock smart home energy with the 2025 Volvo EX90 and dcbel. **Up to $1,300 estimated yearly energy savings by powering your home with your EVs in California.**"[15]
- **Export Rate Potential**: At the time of the original Advice Letter filing, SCE had proposed a new export compensation rate, the Vehicle Grid Resource Proposal ("VGRP"). However, stakeholders were still providing input on the proposal. In September, parties came to a VGRP settlement agreement, which outlines a comprehensive export compensation rate to allow customers to provide exports to the grid based on a static pricing schedule. VGRP may initially enroll up to 100,000 residential customers and will allow unlimited enrollment from non-residential customers. Additionally, SCE is currently designing its dynamic rates for Load Management Standard ("LMS") compliance, which can include export compensation, which is referenced in the original Advice Letter.[16] While the Draft Resolution states that the Dynamic Rate pilot will not have sufficient funds for bidirectional ORCHARD participants,[17] the LMS Dynamic Rate will be an optional rate open to all customers that can enroll ORCHARD participants. While both rates are still pending

[12] SCE Advice Letter at 29
[13] SCE Advice Letter at 23
[14] Bill Crider. *Your Next Side Hustle Might Come from Your Parked Electric Vehicle.* Ford From the Road. October 28, 2025. https://www.fromtheroad.ford.com/us/en/articles/2025/introducing-ford-home-power-management
[15] https://volvo.dcbel.energy/. Accessed February 25, 2026.
[16] SCE Advice Letter at 12.
[17] Draft Resolution at 16.

Commission approval, VGIC expects a decision on VGRP in Spring 2026, which will provide sufficient time for ORCHARD customers to participate in the rate.

- **Demand Response Participation:** Bidirectional charging customers in California are currently participating in California's Emergency Load Reduction Program ("ELRP") and the Demand Side Grid Support ("DSGS") program. While both offerings are currently limited in timeframe, there is a potential for ELRP to be extended through 2028 and DSGS to be provided additional funding from the Legislature.[18]

While bidirectional compensation options remain limited, active work is being done to expand the compensation options for customers. Instead of removing the bidirectional incentives from ORCHARD, the Commission should work to expand program offerings for these customers in this procedural venue or others, such as the current Dynamic Rate Application (for SCE, A.24-06-014, *et al.*) or the Demand Response proceeding (R.25-09-004). Alternatively, the Commission could authorize ORCHARD funding to be used to provide forms of export compensation for customers, which VGIC suggested in response to the Advice Letter.[19] Yet another option would be to require customers participate in *some* form of export rate or program, whether that be VGRP, LMS-compliant dynamic export rate, ELRP, DSGS, or some other option, in order to be eligible to receive the funding. The Commission could also approve the funding under the condition that there must be a live export opportunity in order for SCE to *offer* the incentive to customers. Regardless of future action, the Commission should authorize SCE's bidirectional incentives to ensure that ORCHARD participants are positioned to participate in these evolving market opportunities and avoid missing the opportunity to stimulate adoption of grid-supportive bidirectional charging solutions, shown to yield grid benefits through the extensive research referenced in Section II.A. above.

C. Bidirectional charging systems are currently being offered to customers for operations in grid parallel mode.

There are a wide variety of EV and charger models that allow customers to operate in bidirectional mode when in parallel with the grid. The Draft Resolution expresses doubts that automotive OEMs would enable bidirectional functionality outside of a backup power use case, but in the past year the number of models with this functionality has increased significantly.

Appendix A summarizes vehicles offering grid-parallel bidirectional functionality, including **24 models on the market today from 14 automakers,** with four additional models from four more automakers announced for upcoming release.[20] In total, the list in Appendix A indicates that there are an estimated 215,465 light-duty vehicles and 1,890 electric school buses in California that offer

[18] See ELRP and DSGS Trailer Bill Language at https://trailerbill.dof.ca.gov/public/trailerBill/pdf/1398 and https://trailerbill.dof.ca.gov/public/trailerBill/pdf/1397, respectively.
[19] VGIC Response to Advice Letter 5536-E at 4.
[20] See the full set of data in Appendix A.

bidirectional charging features. Assuming SCE contains 33% of the state's EVs, an estimated 71,727 such vehicles may already be in SCE's service territory.[21]

Appendix A also discusses available bidirectional EVSE. Notably, nearly every bidirectional charging system currently listed in the light-duty segment is certified to UL 1741 SB.[22] In the medium- and heavy-duty segment, all listed bidirectional charging systems are certified to UL 1741 SB. This certification ensures compliance with IEEE 1547-2018 and facilitates interconnection under Rule 21. Nearly all of the bidirectional charging system models listed in Appendix A have been enabled for bidirectional operation in parallel with the grid under certain California programs, including the Emergency Load Reduction Program, Demand Side Grid Support Program, CEC's REDWDS initiative, and PG&E's Hourly Flex Pricing (via PG&E's Vehicle-to-Everything Pilot).

In order to allow customers to operate their bidirectional charging systems in grid-parallel mode, those features must be enabled, which requires the appropriate hardware and software activation after the appropriate utility interconnection study and agreement process is completed.

The Draft Resolution expresses concerns around "the likelihood of whether customers will want to enable the grid-parallel functionality."[23] However, SCE has surveyed customers and provided empirical evidence of customer acceptance and appetite for "V2G energy exports," with 81% of SCE's surveyed customers expressing at least "a little interested" and 16% of customers stating they are "very interested."[24] This level of customer interest makes SCE's bidirectional enrollment goal of 8,700 in ORCHARD very reasonable. Additionally, VGIC today filed a set of Joint Comments with 36 other stakeholders, including customer groups, indicating interest in grid-parallel bidirectional charging features.

D. Other States Are Actively Incentivizing Grid-Parallel Bidirectional Charging.

California utilities are not alone in seeking to unlock bidirectional charging value. Utilities and policymakers in multiple states are proactively addressing the upfront costs and equipment barriers to enable grid-parallel bidirectional EVs.

- **National Grid's Proposed Enhanced V2G Equipment Incentives:** Pending approval of the Massachusetts Department of Public Utilities, National Grid plans to offer ratepayer-funded enhanced incentives to support grid-parallel bidirectional charging systems, for both residential (up to $2,500) and non-residential customers. Customers receiving the incentives must participate in the ConnectedSolutions program – a demand response program that allows for exports, similar to ELRP or DSGS – to provide value to the grid during times of

[21] *Prepared Testimony of Sylvie Ashford Addressing Southern California Edison's Test Year 2025 General Rate Case Load Growth Investments.* A.23-05-010. TURN-07. Page 11-13. https://docs.cpuc.ca.gov/PublishedDocs/SupDoc/A2305010/7090/526146968.pdf
[22] The Tesla Powershare system, currently available for Cybertruck drivers, is not yet certified to UL 1741 SB.
[23] Draft Resolution at 16.
[24] SCE, *Phase 2 of 2025 General Rate Case Amended Rate Design Proposals* (SCE-04A) at p. 125, lines 10-16, served to A.24-03-019 on August 26, 2024. https://docs.cpuc.ca.gov/PublishedDocs/SupDoc/A2403019/7663/538693971.pdf

stress.[25]

- **Connecticut Innovative Energy Solutions:** The Connecticut Innovative Energy Solutions program, overseen by the Public Utilities Regulatory Authority (funded by Connecticut ratepayers) is supporting residential bidirectional chargers to provide virtual power plant services to the grid.[26] Up to $10,800 in incentives for bidirectional equipment is being provided to help offset the upfront equipment costs.[27]

- **MassCEC V2X Demonstration Program:** This $6 million project covers all upfront costs of bidirectional charging equipment and installation for up to 100 residential, commercial, and school bus customers in Massachusetts.[28]

These programs share a common principle: bidirectional EVs are treated as distributed energy resources capable of delivering incremental reliability and system benefits when enabled for grid-parallel operation. Utilities are not limiting these technologies to backup-only use cases; rather, they are proactively removing barriers and designing pathways for participation in demand response, capacity, and export programs.

SCE's request is in line with these goals. Approving the bidirectional incentive would align California with established utility practice nationwide and ensure that ORCHARD captures the full grid value of EVs consistent with SB 676 and the Commission's broader VGI objectives.

E. SCE's ORCHARD proposal resolves one of the biggest barriers to unlocking additional bidirectional charger adoption and operationalization.

In order to unlock bidirectional charging for grid services in California, key barriers will need to be removed. One of the most significant barriers in California is the high upfront costs specific to California, including the $800 Rule 21 interconnection application fee.

The current Rule 21 interconnection application fee for small bidirectional charging systems is disproportionately high. Notably, this fee significantly exceeds those charged by utilities in other

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[25] Massachusetts Electric Company and Nantucket Electric Company, each d/b/a National Grid, *Pre-Filed Direct Testimony of Julia Gold Christopher Coy and Jenna Gould* (Exhibit NG-EVPP-1), December 30, 2025, Docket No. D.P.U. 25-189 https://fileservice.eea.comacloud.net/V3.1.0/FileService.Api/file//aeijjidej?A3keHyyU6XwgzbntVM/zSemplo6W2bDYt6kv4l+nvFGPcSxI+blU344Khxm+qpOeg0hKFj9M9l/xQR8+/8GqPvdGgrFe6XR6ngIfa80wd3rxFD8G4j981M2Rna9aVTXA

[26] Innovative Energy Solutions, *Cycle 2 Awarded Projects*, https://ct-ies.com/cycle-2-awarded-projects/ Accessed February 25, 2026.

[27] Information from Bidirectional Energy at https://www.bidirectional.energy/learn/connecticut. Accessed February 25, 2026.

[28] Massachusetts Clean Energy Center. *Vehicle-to-Everything Demonstration Projects.* https://www.masscec.com/masscec-focus/clean-transportation/electric-vehicle-charging-infrastructure/vehicle-to-everything-demonstration. Accessed February 25, 2026.

states, in most cases by 800%[29] and greatly exceeds the standard $94 fee charged by SCE for similarly-sized rooftop solar customers (i.e., Net Energy Metering or Net Billing Tariff customers) – which may even use the same inverter model as bidirectional charging customers. These fees have an outsized impact on bidirectional charging systems, which have relatively lower upfront costs compared to stationary energy storage systems, rooftop solar PV, or other DERs interconnecting under Rule 21. Since customers are purchasing EVs for transportation uses, the incremental costs to unlock bidirectional charging features are typically only the cost of the bidirectional charger and interconnection. As a result, the $800 interconnection application fee comprises one of the most significant upfront costs for bidirectional charging systems and, in turn, a significant barrier to widespread bidirectional charging system deployment.

The ORCHARD proposal for an $800 incentive is therefore extremely meaningful to unlocking grid-parallel bidirectional charging features and participation. Low-income customers face greater barriers to adopting distributed energy resources, and SCE's plan to offer up to $7,800 in additional incentives for income-qualified customers to help offset other equipment costs ensures equitable access to bidirectional charging systems and the customer-oriented benefits it brings, like lower total cost of ownership. The Commission should therefore amend the Draft Resolution to approve SCE's proposal.

III. CONCLUSION.

VGIC appreciates the opportunity to submit this response to the Draft Resolution. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen
Zach Woogen
Executive Director
VEHICLE-GRID INTEGRATION COUNCIL

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[29] See *Comments of the Vehicle-Grid Integration Council on Order Instituting Rulemaking to Update Distribution Level Interconnection Rules and Regulations.* R.25-08-004. Filed October 20, 2025. *Attachment A: List of Interconnection Fees by State/Utility*. https://static1.squarespace.com/static/5dcde7af8ed96b403d8aeb70/t/68f6e79ed70c7c16ccb997ca/1761011614259/2025-10-20+VGIC+Comments+on+Rule+21+2025+OIR+and+Attachment+-+FINAL.pdf

Appendix A

Table 1: EV and EVSE on California's roads that can be enabled to operate in bidirectional mode when operating in parallel with the grid

Product	Notes	Estimated Vehicle Sales in California
Nissan LEAF	- MY 2013 or later. - Bidirectional charging enabled with dcbel Ara.[30]	54,304 total registered from 2013 through Q3 2025.[31]
Ford F-150 Lightning Electric	- Bidirectional charging enabled with Ford Charge Station Pro and Home Integration System.[32]	Estimated 35,877 total delivered through Q4 2025.[33] Ford customers cannot do bidirectional charging without the associated hardware, software, and activation.
Chevrolet Bolt EV	- MY 2027. - Bidirectional charging enabled with GM Energy PowerShift.[34] - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB certified inverter.	Estimated 0 delivered through 2025.[35]

[30] dcbel, California Energy Commission. *California Connected Home Rebate.* Accessed January 13, 2026. https://redwds.dcbel.energy/

[31] California Energy Commission (2026). *New_ZEV_Sales_updated_10-13-2025_ada.xlsx.* ZEV and Infrastructure Stats Data. Data last updated October 13, 2025. Retrieved January 16, 2026 from https://www.energy.ca.gov/files/zev-and-infrastructure-stats-data.

[32] Bill Crider. *Your Next Side Hustle Might Come from Your Parked Electric Vehicle.* Ford From the Road. October 28, 2025. https://www.fromtheroad.ford.com/us/en/articles/2025/introducing-ford-home-power-management

[33] Ford. *Q4 2025 Sales Release*. https://s205.q4cdn.com/882619693/files/doc_news/2026/Jan/06/Ford-U-S-Q4-2025-Sales-Release.pdf. *Q4 2024 Sales Release* https://s205.q4cdn.com/882619693/files/doc_news/2025/Jan/03/Ford-U-S-Q4-2024-Sales-Release.pdf. *Q4 2023 Sales Release*. https://s205.q4cdn.com/882619693/files/doc_news/2024/Jan/04/q4-2023-sales-final.pdf. YTD 2022, 2023, 2024, and 2025 Ford F-150 Lightning deliveries total 100,149. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

[34] GM Energy. *V2H-Capable GM EVs.* Accessed January 13, 2026. https://gmenergy.gm.com/

[35] Jameson Dow. *The new $29k 2027 Chevy Bolt is now in dealerships, get it before it's gone again.* January 9, 2026. https://electrek.co/2026/01/09/the-refreshed-2027-chevy-bolt-is-hitting-dealerships-get-it-before-its-gone-again/

Chevrolet Silverado EV	- MY 2024, 2025, and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB certified inverter.	Estimated 6,865 total delivered through Q4 2025.[36]
Chevrolet Equinox EV	- MY 2024, 2025, and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB certified inverter.	Estimated 31,098 total delivered through Q4 2025.[37]
Chevrolet Blazer EV	- MY 2024, 2025, and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB certified inverter.	Estimated 16,561 total delivered through Q4 2025.[38]
GMC Hummer EV SUT	- MY 2026. - Bidirectional charging enabled with GM Energy PowerShift.	Estimated 5,655 total delivered through Q4 2025.[39]

[36] General Motors. *GM U.S. Deliveries for Quarter 4 2025.* https://investor.gm.com/static-files/23adce96-ea26-4a6e-82dd-68a0042df51c. *GM U.S. Deliveries for Quarter 4 2024.* https://investor.gm.com/static-files/e1bb38a4-aaf9-4141-b44a-4c61c1f2ec54. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

[37] General Motors. *GM U.S. Deliveries for Quarter 4 2025.* https://investor.gm.com/static-files/23adce96-ea26-4a6e-82dd-68a0042df51c. *GM U.S. Deliveries for Quarter 4 2024.* https://investor.gm.com/static-files/e1bb38a4-aaf9-4141-b44a-4c61c1f2ec54. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

[38] General Motors. *GM U.S. Deliveries for Quarter 4 2025.* https://investor.gm.com/static-files/23adce96-ea26-4a6e-82dd-68a0042df51c. *GM U.S. Deliveries for Quarter 4 2024.* https://investor.gm.com/static-files/e1bb38a4-aaf9-4141-b44a-4c61c1f2ec54. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

[39] General Motors. *GM U.S. Deliveries for Quarter 4 2025.* https://investor.gm.com/static-files/23adce96-ea26-4a6e-82dd-68a0042df51c. *GM U.S. Deliveries for Quarter 4 2024.* https://investor.gm.com/static-files/e1bb38a4-aaf9-4141-

	- Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB certified inverter.	
GMC Sierra EV	- MY 2024, 2025, and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB certified inverter.	Estimated 3,505 total delivered through Q4 2025.[40]
Cadillac CELESTIQ	- MY 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB certified inverter.	Estimated very few. CELESTIQ first delivered in mid-2025.[41]
Cadillac VISTIQ	- MY 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB certified inverter.	Estimated 2,822 total delivered through Q4 2025.[42]

b44a-4c61c1f2ec54. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

[40] General Motors. *GM U.S. Deliveries for Quarter 4 2025.* https://investor.gm.com/static-files/23adce96-ea26-4a6e-82dd-68a0042df51c. *GM U.S. Deliveries for Quarter 4 2024.* https://investor.gm.com/static-files/e1bb38a4-aaf9-4141-b44a-4c61c1f2ec54. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

[41] GM News. *First Cadillac CELESTIQ delivered, marking a new era for the Standard of the World.* June 24, 2025. https://news.gm.com/home.detail.html/Pages/topic/us/en/2025/jun/0624-First-Cadillac-CELESTIQ-delivered.html

[42] General Motors. *GM U.S. Deliveries for Quarter 4 2025.* https://investor.gm.com/static-files/23adce96-ea26-4a6e-82dd-68a0042df51c. *GM U.S. Deliveries for Quarter 4 2024.* https://investor.gm.com/static-files/e1bb38a4-aaf9-4141-b44a-4c61c1f2ec54. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

Cadillac ESCALADE IQ	- MY 2025 and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB certified inverter.	Estimated 3,146 total delivered through Q4 2025.[43]
Cadillac OPTIQ	- MY 2025 and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB certified inverter.	Estimated 4,365 total delivered through Q4 2025.[44]
Cadillac LYRIQ	- MY 2024, 2025, and 2026. - Bidirectional charging enabled with GM Energy PowerShift. - Although GM does not explicitly advertise grid-parallel bidirectional charging, the GM Energy V2X Enablement Kit does include a UL 1741 SB certified inverter.	Estimated 20,964 total delivered through Q4 2025.[45]

[43] General Motors. *GM U.S. Deliveries for Quarter 4 2025.* https://investor.gm.com/static-files/23adce96-ea26-4a6e-82dd-68a0042df51c. *GM U.S. Deliveries for Quarter 4 2024.* https://investor.gm.com/static-files/e1bb38a4-aaf9-4141-b44a-4c61c1f2ec54. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

[44] General Motors. *GM U.S. Deliveries for Quarter 4 2025.* https://investor.gm.com/static-files/23adce96-ea26-4a6e-82dd-68a0042df51c. *GM U.S. Deliveries for Quarter 4 2024.* https://investor.gm.com/static-files/e1bb38a4-aaf9-4141-b44a-4c61c1f2ec54. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

[45] General Motors. *GM U.S. Deliveries for Quarter 4 2025.* https://investor.gm.com/static-files/23adce96-ea26-4a6e-82dd-68a0042df51c. *GM U.S. Deliveries for Quarter 4 2024.* https://investor.gm.com/static-files/e1bb38a4-aaf9-4141-b44a-4c61c1f2ec54. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

Tesla Cybertruck	- Bidirectional charging enabled with Tesla Universal Wall Connector.[46]	14,360 total registered through Q3 2025.[47]
Kia EV9	- Bidirectional charging enabled with Wallbox Quasar 2.[48]	Estimated 13,277 total delivered through Q4 2025.[49]
Volvo EX90	- Bidirectional charging enabled with dcbel Ara.[50]	Estimated 1,708 total delivered through Q4 2025.[51]
Polestar 3	- Bidirectional charging enabled with dcbel Ara.[52]	958 total registered through Q3 2025.[53]
BlueBird	- 155 kWh or 196 kWh school buses[54]	915 through Q4 2024.[55]

[46] Tesla. *Powershare.* Accessed January 13, 2026. https://www.tesla.com/powershare

[47] California Energy Commission (2026). *New_ZEV_Sales_updated_10-13-2025_ada.xlsx.* ZEV and Infrastructure Stats Data. Data last updated October 13, 2025. Retrieved January 16, 2026 from https://www.energy.ca.gov/files/zev-and-infrastructure-stats-data.

[48] Kia EV9 x Wallbox Quasar 2. Accessed January 13, 2026. https://wallbox.com/en_us/kia-ev9-quasar-2

[49] Best-Selling Cars. *2025 (Full Year) USA: Kia America US Car Sales by Model.* https://www.best-selling-cars.com/usa/2025-full-year-usa-kia-america-us-car-sales-by-model/. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

[50] dcbel, California Energy Commission. *California Connected Home Rebate.* Accessed January 13, 2026. https://redwds.dcbel.energy/

[51] Volvo Cars. *Sales Volumes: United States – Retail Sales by Car Model – December 2025.* Accessed January 16, 2026. https://www.volvocars.com/us/media/corporate/sales-volumes/. Assumes California comprises 35.82% of all vehicle deliveries, based on Veloz's *California EV Market Report*, accessed January 16, 2026, listing 2,468,158 CA EV sales and 6,889,728 U.S. EV Sales. https://www.veloz.org/ev-market-report/

[52] dcbel, California Energy Commission. *California Connected Home Rebate.* Accessed January 13, 2026. https://redwds.dcbel.energy/

[53] California Energy Commission (2026). *New_ZEV_Sales_updated_10-13-2025_ada.xlsx.* ZEV and Infrastructure Stats Data. Data last updated October 13, 2025. Retrieved January 16, 2026 from https://www.energy.ca.gov/files/zev-and-infrastructure-stats-data.

[54] BusinessWire. *Blue Bird Receives Record Order of 180 Electric School Buses.* January 16, 2024. https://www.businesswire.com/news/home/20240116715215/en/Blue-Bird-Receives-Record-Order-of-180-Electric-School-Buses

[55] California Energy Commission (2026). *Medium_Heavy_Duty_Vehicle_Population_Last_updated_04-30-2025_ada.xlsx.* ZEV and Infrastructure Stats Data. Data last updated April 30, 2025. Retrieved January 16, 2026 from https://www.energy.ca.gov/files/zev-and-infrastructure-stats-data.

BYD / RIDE	- 156 kWh, 230 kWh, or 288 kWh school buses[56]	127 through Q4 2024.[57]
IC Bus	- 210 kWh or 315 kWh school buses[58]	731 through Q4 2024.[59]
Green Power Motors	- 118 kWh or 193 kWh school buses[60]	117 through Q4 2024.[61]
Thomas Built Buses	- 226 kWh school buses[62]	Unknown
Tellus Power Green	- 20, 30, 40, or 60 kW EVSE[63]	Unknown
InCharge	- 22, 44, or 66 kW EVSE[64]	Unknown
Heliox	- 60 kW EVSE[65]	Unknown

In addition to the above-listed currently available solutions, the following EVs are associated with announced, but not yet available, bidirectional charging features: Hyundai Ioniq 9,[66] Kia EV6,[67] Acura RSX,[68] and Rivian R2.[69] EVSE manufacturers that have announced but not yet released

[56] PG&E. *Vehicle-to-Everything (V2X) pilot program: Eligible Product List for the V2X Commercial Pilot.* https://www.pge.com/en/clean-energy/electric-vehicles/getting-started-with-electric-vehicles/vehicle-to-everything-v2x-pilot-programs.html. Retrieved January 13, 2026.

[57] California Energy Commission (2026). *Medium_Heavy_Duty_Vehicle_Population_Last_updated_04-30-2025_ada.xlsx.* ZEV and Infrastructure Stats Data. Data last updated April 30, 2025. Retrieved January 16, 2026 from https://www.energy.ca.gov/files/zev-and-infrastructure-stats-data.

[58] IC Bus. *Electric CE Series School Bus.* https://www.icbus.com/-/media/Project/Navistar/ICBus/ICBus/electric_ce_series_brochure_spread_format.pdf. Retrieved January 7, 2025.

[59] California Energy Commission (2026). *Medium_Heavy_Duty_Vehicle_Population_Last_updated_04-30-2025_ada.xlsx.* ZEV and Infrastructure Stats Data. Data last updated April 30, 2025. Retrieved January 16, 2026 from https://www.energy.ca.gov/files/zev-and-infrastructure-stats-data.

[60] California HVIP. *School Bus: Vehicle Type – ZESBI $ Eligible.* https://californiahvip.org/vehicle-category/school-bus/?type=655. Retrieved January 7, 2025.

[61] California Energy Commission (2026). *Medium_Heavy_Duty_Vehicle_Population_Last_updated_04-30-2025_ada.xlsx.* ZEV and Infrastructure Stats Data. Data last updated April 30, 2025. Retrieved January 16, 2026 from https://www.energy.ca.gov/files/zev-and-infrastructure-stats-data.

[62] Thomas Built Buses, Retrieved March 3, 2026 from https://thomasbuiltbuses.com/electric-school-buses/utility-providers/

[63] PG&E. *Vehicle-to-Everything (V2X) pilot program: Eligible Product List for the V2X Commercial Pilot.* https://www.pge.com/en/clean-energy/electric-vehicles/getting-started-with-electric-vehicles/vehicle-to-everything-v2x-pilot-programs.html. Retrieved January 7, 2025.

[64] InCharge. *Chargers & Hardware.* https://inchargeus.com/chargers/. Retrieved January 7, 2025.

[65] Heliox. *Heliox 60 kW*. https://www.heliox-energy.com/us-products/60-kw-dc-charger. Retrieved January 7, 2025.

[66] Hyundai. *Hyundai Motor Group Expands EV Energy Services with Vehicle to Grid and Vehicle to Home*. November 28, 2025. https://www.hyundainews.com/releases/4634

[67] *Ibid.*

[68] Honda. *Honda Highlights Home and Vehicle Energy Management Technologies at RE+ 25*. September 3, 2025. https://hondanews.com/en-US/releases/honda-highlights-home-and-vehicle-energy-management-technologies-at-re-25

[69] EV.com. *$45,000 Rivian R2 Will Power Homes With New Bidirectional Charging Technology.* November 6, 2025. https://ev.com/news/45000-rivian-r2-will-power-homes-with-new-bidirectional-charging-technology

bidirectional charging solutions include: Emporia,[70] ChargePoint,[71] Enphase,[72] SolarEdge, and Autel.

[70] Emporia. *Coming Soon: V2X Bi-Directional Charger.* Accessed January 13, 2026. https://www.emporiaenergy.com/how-the-emporia-v2x-charger-works/

[71] ChargePoint. *ChargePoint and Eaton launch breakthrough ultrafast DC V2G chargers and power infrastructure to accelerate the future of EV charging*. August 28, 2025. https://www.chargepoint.com/about/news/chargepoint-and-eaton-launch-breakthrough-ultrafast-dc-v2g-chargers-and-power

[72] Enphase. *IQ Bidirectional EV Charger.* Accessed January 13, 2026. https://enphase.com/ev-chargers/bidirectional